

Abdul Rafay Imran

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Summary

Software Engineering student specializing in full-stack systems, applied AI, and Web3 protocol design. Built and shipped 18+ production-grade projects including zero-knowledge payment protocols, enterprise trust platforms, cybersecurity defense systems, and clinical AI decision-support tools. Experienced across React, Next.js, Node.js, FastAPI, Hedera, Starknet, and ML pipelines including scikit-learn, XGBoost, and Transformers. Strong foundation in system design, security engineering, and scalable architecture.

Technical Skills

Programming Languages: JavaScript, TypeScript, Python, Solidity, Cairo, PHP, SQL
Frontend Development: React.js, Next.js, Angular, Tailwind CSS, Framer Motion, Vite, HTML5, CSS3
Backend Development: Node.js, Express.js, FastAPI, REST APIs, WebSockets, JWT, OAuth, RBAC
Blockchain & Web3: Hedera, Starknet, EVM, ZK-Proofs, HTLCs, Atomic Swaps, DeFi Protocols
AI & Machine Learning: scikit-learn, XGBoost, Random Forest, Transformers, NLP, MedGemma, Pandas, NumPy
Databases: MongoDB, MySQL, Redis
Tools & Practices: Docker, Git, GitHub, CI/CD, Agile/Scrum, Technical Documentation

Education

University of Karachi <i>Bachelor of Science in Software Engineering</i>	2023 – Present Karachi, Pakistan
Aptech Learning <i>Diploma in Software Engineering</i>	2022 – 2025 Karachi, Pakistan

Selected Projects – Full portfolio: github.com/RafayImraan

ShadowPay – *Privacy-Preserving Bitcoin Payment Protocol* | Cairo, Starknet, Next.js, ZK-Proofs GitHub
– Designed a zero-knowledge payment layer on Starknet L2 using Pedersen commitments and nullifier trees to enable confidential Bitcoin transactions with cryptographic verifiability.
– Built Cairo smart contracts for proof verification, commitment storage, and double-spend prevention, integrated with a Next.js frontend and wallet connection flow.
– Implemented a privacy-preserving transaction system where payment details remain hidden while proofs stay publicly verifiable on-chain.

Reality Check – *Explainable AI Manipulation Detector* | React, FastAPI, Transformers, NLP GitHub
– Engineered a hybrid rule-based and transformer pipeline to detect manipulation patterns including fear, urgency, authority abuse, polarization, and missing evidence.
– Implemented confidence scoring, fairness auditing, claim-versus-evidence validation, and exportable compliance reports for transparent AI analysis.
– Delivered a full-stack platform with analysis workspace, risk dashboard, simulations, and batch processing API for scalable text review.

FrostByte – *AI Cybersecurity Defense System* | Next.js, FastAPI, Chrome Extension GitHub
– Built a multi-layer threat detection engine combining machine learning, threat intelligence, and heuristic analysis to identify phishing, scams, and malicious URLs in real time.
– Developed a browser extension that performs live page scanning, highlights suspicious elements, shows warnings, and blocks high-risk content.
– Integrated analytics dashboard, monitoring history, and configurable alerting for deployable cybersecurity operations.

TrustNet AI – *B2B Trust Coordination Platform* | React, FastAPI, Node.js, Hedera GitHub
– Architected an AI-driven trust decision system for onboarding, payouts, and risk checks using explainable workflows and structured routing.
– Anchored trust decisions on Hedera to create immutable audit logs, portable trust history, and token-based incentive mechanisms.
– Built enterprise-ready features including multi-tenant workspaces, role-based controls, webhook integrations, and compliance monitoring tools.

GridMind AI – *Urban Infrastructure Failure Predictor* | FastAPI, scikit-learn, React GitHub
– Developed a hybrid prediction engine combining rule-based cascading-failure logic with machine learning to forecast infrastructure risks across power, water, traffic, and transformers.
– Generated explainable risk reports with confidence levels, ranked contributing factors, scenario simulations, and weather anomaly correlation.
– Delivered a monitoring platform with real-time analysis, historical tracking, and offline fallback support for uninterrupted critical system evaluation.

MedAssist – *Offline Clinical Triage AI* | React, FastAPI, MedGemma, Docker GitHub
– Built an offline-capable clinical triage system that analyzes symptoms, vitals, and medical history to generate structured triage summaries, urgency levels, red flags, and care recommendations.

- Implemented dual-path inference using MedGemma with deterministic fallback logic to ensure safe and schema-consistent outputs under varying hardware constraints.
 - Containerized the application with Docker and configurable runtime selection for local, CPU, or GPU-based execution.
- GBF** – *Green Bond Fractionalization DeFi Protocol* | *Hedera, React, Node.js* GitHub
- Designed a DeFi platform that fractionalizes institutional green bonds into low-cost units using Hedera to broaden investor access.
 - Paired investment flows with verifiable ecological impact evidence to support transparent and traceable sustainability reporting.
- Medicare** – *Hospital Management System* | *MongoDB, Express.js, React.js, Node.js* GitHub
- Developed a full-stack healthcare platform with role-based access for admins, doctors, and patients, covering appointments, billing, telehealth, feedback, and dashboards.
 - Implemented JWT authentication, hierarchical RBAC, caching, rate limiting, notifications, and customizable analytics views for responsive cross-device use.
- BSwap** – *Cross-Chain Atomic Swap Protocol* | *Next.js, Bitcoin Testnet, Hedera Testnet, HTLC* GitHub
- Implemented a trustless cross-chain atomic swap between Bitcoin and Hedera using SHA-256 hash time-locked contracts with enforced timelock ordering.
 - Built a non-custodial coordination API and dashboard for swap initiation, verification, claim execution, and refund monitoring.
- ARIA** – *Interactive Narrative Simulation Engine* | *React, TypeScript, Vite, Tailwind CSS* GitHub
- Engineered a branching narrative simulation with live audience voting, dynamic city metrics, and decision-based progression.
 - Designed extensible architecture with configurable voting windows, modular story flow, and immersive presentation support.

Additional Projects

Forus <i>Confidential EVM fund-receiving protocol using stealth address patterns</i>	GitHub
Veluxe Motors <i>Premium car dealership website with Next.js and lead generation optimization</i>	GitHub
Portfolio <i>Next.js developer portfolio with Framer Motion, theme switching, and AI assistant widget</i>	GitHub
GitHub Time Machine <i>Interactive repository history explorer built with React and GitHub REST API</i>	GitHub
Weather Analysis <i>Predictive weather modeling with XGBoost, Random Forest, and interactive dashboard</i>	GitHub
Crop Yield Prediction <i>Agricultural yield analysis for Pakistan using machine learning models</i>	GitHub
Employment Trend Analysis <i>Pre- and post-COVID unemployment trend analysis with visualizations</i>	GitHub
Car Price Prediction <i>Linear Regression pipeline with feature engineering and model validation</i>	GitHub

Achievements

First Prize , Speed Programming Competition – <i>Aptech Learning, Karachi, Pakistan</i>	2023
Second Prize , Web Designing Competition – <i>Aptech Learning, Karachi, Pakistan</i>	2023

Additional

Methodologies: Agile/Scrum, Test-Driven Development, CI/CD, Code Review, Technical Documentation

Soft Skills: Cross-Functional Communication, Technical Leadership, Analytical Problem-Solving, Rapid Prototyping

Languages: English (Professional Fluency), Urdu (Native Speaker)